THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 3: **Metals and Non-metals** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An element is very soft, can be cut with a knife and reacts vigorously with cold water. It is: 1

(a) copper **(b)** potassium **(c)** lead **(d)** silver

2. The non-metal that is a liquid at room temperature is: 1

(a) mercury **(b)** chlorine **(c)** bromine **(d)** iodine

3. The best conductors of heat among metals are: 1

(a) iron and zinc **(b)** silver and copper

(c) sodium and potassium **(d)** lead and mercury

4. A student states: “All metals are solids at room temperature.” The correct response is: 1

(a) Correct **(b)** Correct except iron

(c) Incorrect — mercury is a liquid metal **(d)** Cannot be determined

5. For Al_2O_3, Student A says it is amphoteric; Student B says it is purely basic. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. Silver articles turn black in air due to the formation of: 1

(a) silver nitrate **(b)** silver oxide **(c)** silver chloride **(d)** silver sulphide

7. Which metal is stored under kerosene? 1

(a) copper **(b)** aluminium **(c)** iron **(d)** sodium

8. The process of coating iron with zinc is called: 1

(a) electroplating with tin **(b)** alloying

(c) galvanisation **(d)** anodising

9. Aqua regia is a mixture of conc. HCl and conc. HNO_3 in the ratio: 1

(a) 1 : 3 **(b)** 2 : 1 **(c)** 3 : 1 **(d)** 1 : 1

10. Ionic compounds generally have: 1

(a) a liquid state at room temperature

(b) low boiling points

(c) high melting points

(d) no electrical conductivity in molten state

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is sodium kept immersed in kerosene oil? 2

12. Name a metal and a non-metal that exist as liquids at room temperature. 2

13. What is an alloy? Give one example with its constituents. 2

14. A student says that hydrogen gas is evolved when any metal reacts with nitric acid. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Show the formation of NaCl and $MgCl_2$ by the transfer of electrons. 3

16. What are amphoteric oxides? Give two examples and write the equations of one with an acid and a base. 3

17. Write equations for the reactions of 3
(i) iron with steam
(ii) calcium with water
(iii) potassium with water.

OR

Give reasons: 3
(i) Platinum, gold and silver are used to make jewellery.
(ii) Sodium, potassium and lithium are stored under oil.
(iii) Aluminium is used to make cooking utensils.

18. What is the thermit reaction? Write its equation and one use. 3

OR

Explain why ionic compounds 3
(i) conduct electricity in molten state
(ii) have high melting points
(iii) are soluble in water but not in kerosene.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is the reactivity series? 5
(ii) Explain the extraction of a metal of medium reactivity (e.g. zinc) from its sulphide ore with equations.
(iii) Why are carbonate and sulphide ores converted to oxides?

OR

(i) What is electrolytic refining? 5
(ii) Draw/describe the set-up for refining copper.
(iii) What are the anode, cathode and electrolyte?
(iv) Where does anode mud collect?

20. **Case study:** Metal X is found in nature in the free state. Metal Y burns in air with a dazzling white flame. Metal Z is extracted by electrolysis of its molten chloride. 5
(i) Name X, Y and Z (one example each).
(ii) Write the equation for Y burning.
(iii) Why can Z not be extracted by reduction with carbon?

OR

(i) Compare metals and non-metals on the basis of 5
(a) physical state
(b) lustre
(c) conductivity
(d) nature of oxides.
(ii) Name one non-metal that is lustrous and one that conducts electricity.

END OF QUESTION PAPER — **Total: 40 Marks**

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